

ASYNCHRONOUS CODE

Asynchronous

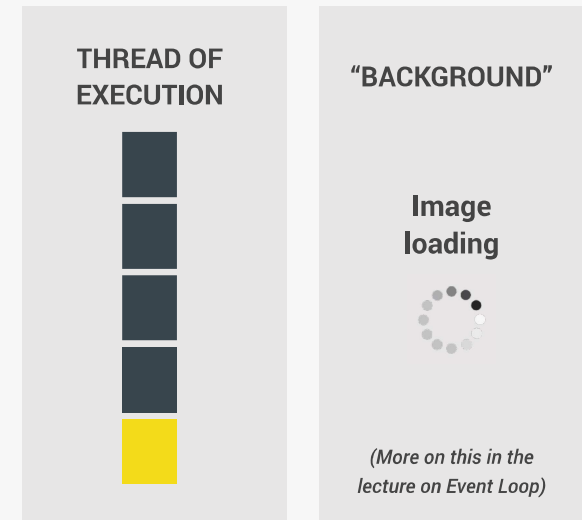
```
const img = document.querySelector('.dog');
img.src = 'dog.jpg';
img.addEventListener('load', function () {
  img.classList.add('fadeIn');
});
p.style.width = '300px';
```

CALLBACK WILL RUN AFTER IMAGE LOADS

Example: Asynchronous image loading with event and callback

Other examples: Geolocation API or AJAX calls

addEventListener does NOT automatically make code asynchronous!



ASYNCHRONOUS

Coordinating behavior of a program over a period of time

- Asynchronous code is executed **after** a task that runs in the "background" finishes;
- Asynchronous code is **non-blocking**;
- Execution doesn't wait for an asynchronous task to finish its work;
- Callback functions alone do **NOT** make code asynchronous!

WHAT ARE AJAX CALLS?

AJAX

Asynchronous JavaScript And XML: Allows us to communicate with remote web servers in an **asynchronous way**. With AJAX calls, we can **request data** from web servers dynamically.



WHAT IS AN API?

API

👉 **Application Programming Interface:** Piece of software that can be used by another piece of software, in order to allow **applications to talk to each other**;

👉 There are be many types of APIs in web development:

DOM API

Geolocation API

Own Class API

"Online" API

Just "API"

👉 **"Online" API:** Application running on a server, that receives requests for data, and sends data back as response;

👉 We can build **our own** web APIs (requires back-end development, e.g. with node.js) or use **3rd-party** APIs.



AJAX

~~XML~~

~~XML data
format~~



JSON data
format

```
{  
  "publisher": "101 Cookbooks",  
  "title": "Best Pizza Dough Ever",  
  "source_url": "http://www.101cookbo",  
  "recipe_id": "47746",  
  "image_url": "http://forkify-api.he",  
  "social_rank": 100,  
  "publisher_url": "http://www.101coo",  
}
```

Most popular
API data format

There is an API for
everything

- 👉 Weather data
- 👉 Data about countries
- 👉 Flights data
- 👉 Currency conversion data
- 👉 APIs for sending email or SMS
- 👉 Google Maps
- 👉 Millions of possibilities...



